

Caleb Poggemeyer

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Education

University of Nebraska–Lincoln

Expected May 2026

B.S. in Computer Engineering (*AI Emphasis*)

GPA: 3.52

- Nebraska Regents Scholarship Recipient; Dean's List

Technical Skills

Languages: C, C#, Java, Python, SQL, Assembly, VHDL

Frameworks/Technologies: React, Axios, .NET 8, ASP.NET Web API, Entity Framework Core, TensorFlow, Keras, Hugging Face, scikit-learn

Data/ML Tools: NumPy, pandas, SciPy, Matplotlib, Seaborn

Focus Areas: AI Engineering, Prompt Engineering, Full-Stack Development, Embedded Systems

Tools: Git, GitHub Actions, Pytest, xUnit, Arduino IDE, Linux/Unix, Jupyter Notebook, VS Code

Projects

Senior Design Project — Squad Lead

Embedded Systems, Arduino

- Leading a six-person team developing an industry-sponsored automated air impact wrench with integrated paint-marking for CLAAS.
- Designing the embedded control system integrating vibration sensing, RPM detection, relay control, and automated paint application.
- Coordinating hardware/software integration, prototype testing, and weekly technical reviews with the sponsor.

Prompt Forge — Prompt Engineering Platform (In Progress)

React, .NET 8, EF Core

- Developing a full-stack AI platform using React 19 (Axios), .NET 8 Web API, EF Core, and SQLite that guides users through a structured interview to generate optimized prompts for Claude, Gemini, and Llama 3.
- Built a dual-track LLM evaluation pipeline combining human ratings with an automated Gemini “LLM-as-Judge” scorer using a shared seven-dimension rubric and exponential backoff for reliable API pacing.
- Designed a 108-prompt benchmark with professional and learner evaluators to measure quality and inter-rater agreement, backed by 89 xUnit tests and a CSV ingestion pipeline for evaluation data.

Full-Stack Web Application

React, C#, SQL

- Built a web application with search functionality using React, C#/.NET, and SQL.
- Collaborated in an Agile team to implement features, integrate backend APIs, and debug frontend/backend interactions.
- Implemented CI workflows with GitHub Actions to automate testing and ensure code quality.

Price Comparison Application

C#, REST APIs

- Built a backend application that queries the Amazon API and parses NET32 JSON data to compare product pricing across vendors.
- Implemented data parsing and normalization pipelines to standardize pricing from multiple sources.
- Designed the application with a modular backend architecture to support future frontend integration.

Experience

Nebraska City Dance Studio — Audio Director

Summers 2021–2025

- Managed live sound production for annual performances with 100+ attendees.
- Diagnosed and resolved technical issues under time-critical conditions while coordinating with instructors.

City of Nebraska City — Summer Tennis Director

Summers 2021–2024

- Led a youth tennis program serving 30+ participants weekly while maintaining a safe learning environment.
- Adapted lesson plans and schedules to accommodate varying skill levels.

Leadership & Involvement

Eagle Scout — Boy Scouts of America

2009–2021

Vice President — UNL Intramural Broomball Team

2022–Present

Member — UNL Intramural Tennis Team

2022–Present